



Capstone Project 2

CMU-SE 451

Architecture Document

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DnD Translate - Sign Language

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Revision history

Version	Date	Description of Changes	Author
1.0	15/02/2024	Initial document	Team

1. Project Overview**PROJECT INFORMATION**

Project acronym	SLDS		
Project Title	Sign Language Detection System		
Start Date	15 - Feb - 2024	End Date	20 - May - 2024
Lead Institution	International School, Duy Tan University		
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1. Introduction

1.1. Project overview

Project definition

This project aims to develop an online system that supports communication via video calls between sign language users and normal people. The system will use real-time translation and gesture recognition technology to convert sign language into spoken language and vice versa, facilitating remote communication.

Business needs/ User needs

- Problems and Limitations
 - Communication is an indispensable part of everyday life, but for deaf and mute people who use sign language, this is often difficult.
 - Limited remote access and communication add to their barriers, especially as current technologies often do not fully meet their needs..
- Why should this project be done?
 - Effective communication tools that bridge the gap between sign language and spoken language.
 - Real-time translation capabilities to facilitate seamless interaction with hearing individuals.
 - User-friendly interface and intuitive design for easy navigation and operation.
Compatibility with different devices and platforms to ensure accessibility.
 - Reliable and accurate translation algorithms that capture the nuances of sign language gestures and spoken language expressions.

1.2. Scope and Purpose

1.2.1. Scope

- The project focuses on developing a video call communication system between sign language users and normal people.

- The scope of the project includes building the user interface, integrating real-time translation and gesture recognition technology, as well as ensuring system flexibility and security.

1.2.2. Purpose

- The main aim of the project is to create an effective communication tool between two groups of people with different communication approaches: sign language users and normal people.
- The project aims to reduce communication barriers and enhance the connection and integration of the deaf and mute community into society through technology.

1.3. Business driver

Business Problems:

- Managing electrical devices in the home can be cumbersome and time-consuming, especially for people with disabilities or mobility issues.
- Electrical hazards, such as fires and electrical shocks, are a serious safety concern.
- Traditional home security systems can be expensive and difficult to install.

Business Needs:

- To develop a product that is affordable and accessible to a wide range of consumers.
- To create a product that is easy to install and use, even for people with limited technical expertise.
- To develop a product that provides a comprehensive solution for managing electrical devices and improving home safety.

2. Architecture driver

2.1. Business constraints

- Resource: 4 people.
- Budget: \$3828.
- Time: The project must be completed within 4 months.

2.2. Technical constraints

The technology for development is Android Studio, which is the tool for Application development and the official IDE for the Application platform.

Technical to develop

- Language: Python, Javascript
- Operating system: Microsoft Windows
- Develop tools: Visual Studio Code, Pycharm
- Database Management System: Postgresql

Environment

- Tool: Web
- Operating systems: Microsoft Windows

Other Constraints

- Resource: 4 people
- Budget: Limited
- Time: The project must be completed within 3 months
- Area: Duy Tan University

2.3. Functional requirement

References to Product Backlog specification of ProductBacklogV1.0.docx

2.4. Quality attributes

2.4.1. Utility table

There are the following quality attributes that drive the design of architecture. Each quality attribute scenario is ranked with importance (I) defined by the Product Owner, and the estimated level of difficulty (D). Both values are based on a scale of High (H) - Medium (M) - Low (L).

2.4.2. Quality attributes

2.4.2.1. Security

Scenario: A user registers an account on the website.	
Type	Security
Stimulus	Registration attempt
Source of stimulus	User
Environment	Website operational
Artifact stimulated	System
Response	Encrypt password
Response measure	Encrypted through 2 layers of security

Performance

Scenario: A user uploads a video for sign language translation.	
Type	Performance
Stimulus	Video upload
Source of stimulus	User
Environment	Normal website operation
Artifact stimulated	Translation engine
Response	Translate sign language in real-time
Response measure	Less than 5 seconds

Scenario: When a user requests to update account information, the system will update account information and respond within 7 seconds with a stable network connection.	
Type	Performance
Stimulus	Update account information
Source of stimulus	User, Project owner
Environment	In runtime and stable network connection
Artifact stimulated	System
Response	Information has been updated
Response measure	Within 7 seconds

2.4.2.2. Usability

Scenario: A new user navigates the website to use the translation service.	
Type	Usability
Stimulus	Navigation by user
Source of stimulus	User, Project owner
Environment	System
Artifact stimulated	Website navigation elements
Response	Intuitive navigation and clear instructions

Response measure	User successfully starts translation within 3 clicks
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2.4.2.3. Modifiability

Scenario: The development team needs to update the translation algorithm.	
Type	Modifiability
Stimulus	Update request
Source of stimulus	Development team
Environment	Development environment
Artifact stimulated	Translation algorithm module
Response	Update algorithm without system downtime
Response measure	Update deployed within 1 hour with zero downtime

3. Architecture overview

This section shows the diagrams that bound our target system and describes the architecture and interaction between components

3.1. System context

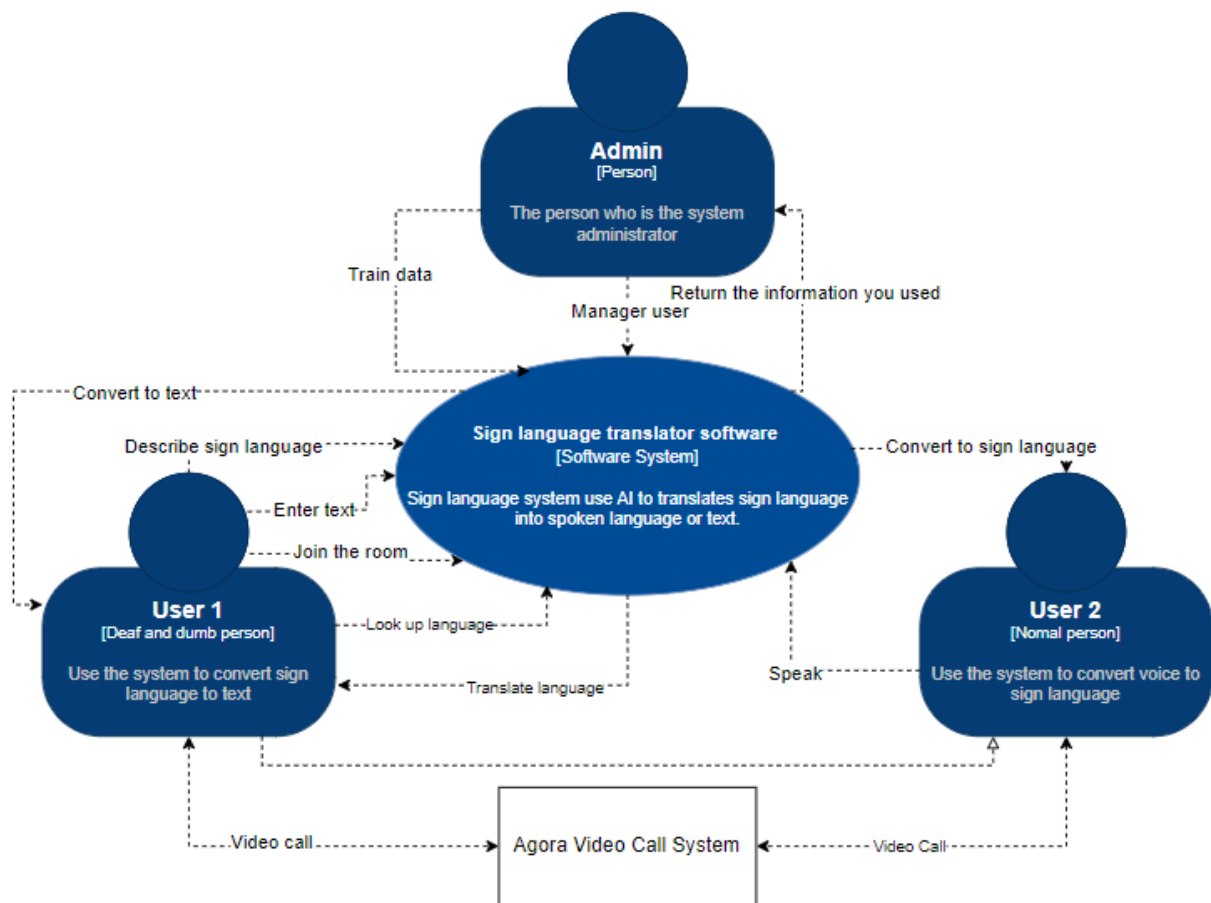


Figure 1: System Context Overview

Users:

- Join the room
- Describe language
- Look up language

Sign language system:

- Convert language
- Learning data use provide

Admin:

- Pass data for training
- Manage users

Agora Video Call System:

- Video call

3.2. Component and connector

We mainly used a C&C view to argue and reason about architectural properties, quality attribute requirements, and functional requirements that the system must add here.

This view type partitions the system into components that have some runtime presence such as processes, objects, data stores, and connectors, or that represent pathways of communication such as data flows and access to shared storage

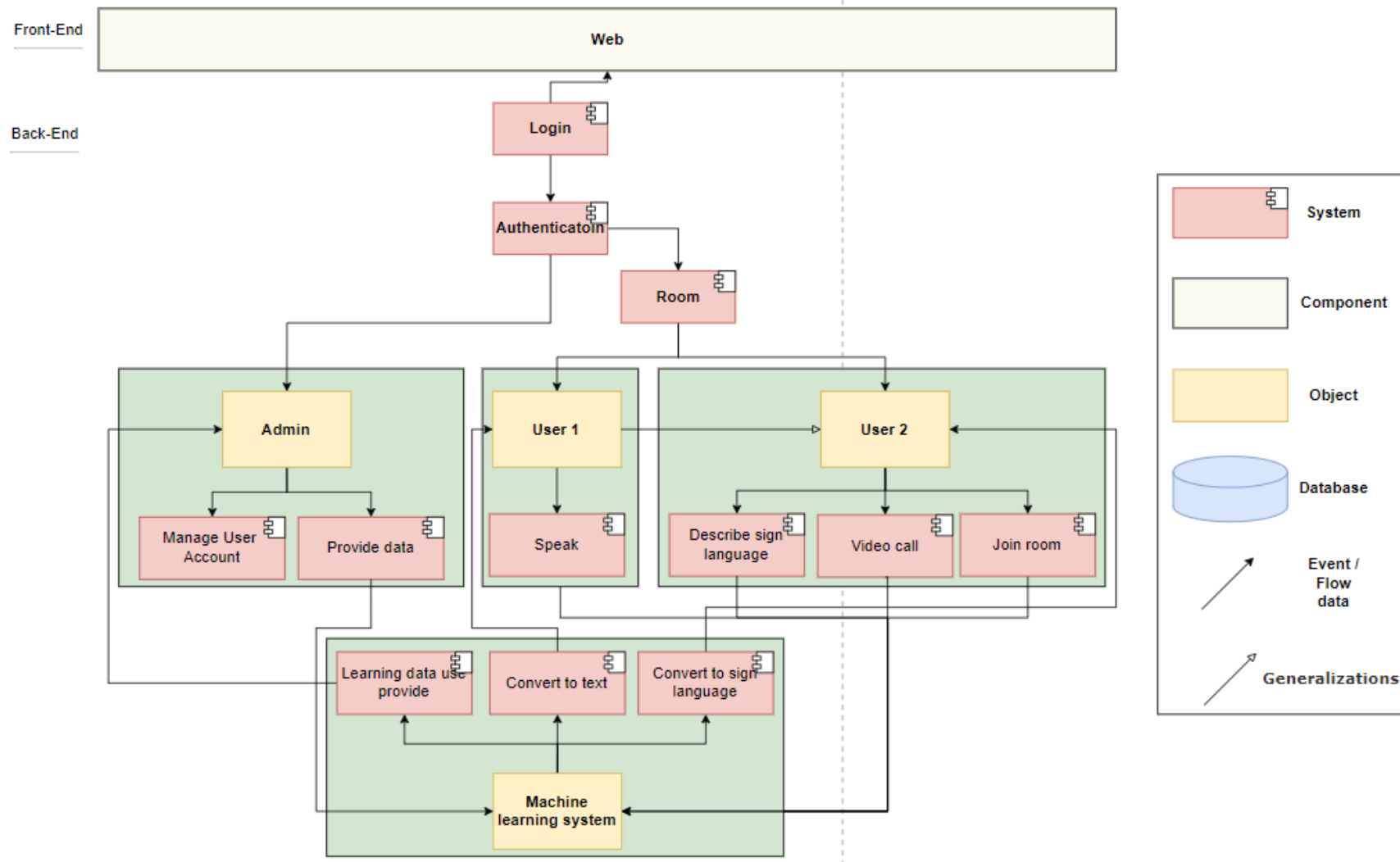


Figure 2: C&C

Users can interact with the system through a mobile application or a web interface to control the connected device. Control data is transmitted from the application to the ESP8266 device over the Internet, which then forwards it to Firebase for storage and processing. Upon any changes in the device's state, Firebase notifies the ESP8266, and the device updates its state accordingly. This process facilitates flexible and efficient remote control of the device.

Element	Responsibilities
Login/Register	Users can create an account or log in to the system.
User Interface	Provides a deaf-friendly interface, including sign language and text.
Search Function	Allows admin to search for information or other users easily.
Machine Learning	Uses machine learning technology to convert voice to text and sign language, making communication smoother.
User Interaction	Users can communicate with each other through text or sign language.
Multi-Language Support	The system supports many different languages, enhancing accessibility.
Security	Ensuring the safety of users' personal information and communication data.

3.3. Module view

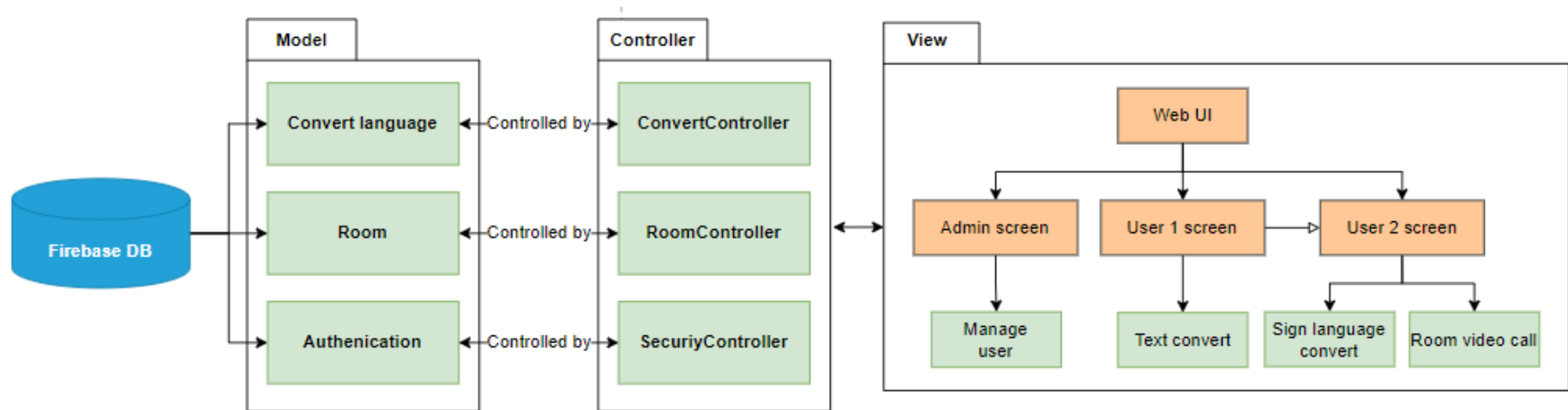


Figure 2: *Module View*

Prose

Element	Responsibilities
Web View	Provides a user-friendly and easy-to-use web interface. To support communication for people with disabilities to the maximum extent.
Models	Object data modules connect to tables in the database for transmission to the system interface.
Controller	Manage the flow of information between applications and data models. Handles requests from users through the application interface and passes control to the corresponding model.

Description: This system aims to bring convenience and efficiency in managing home electrical systems through the use of mobile phones. Users can control and monitor power consumption, schedule operations, and automate electrical appliances in the home, creating a safe, economical, and environmentally friendly digital environment.

3.4. Allocation view

The allocation view models the run-time architecture of a system. It shows the configuration of the hardware elements when the system is deployed.

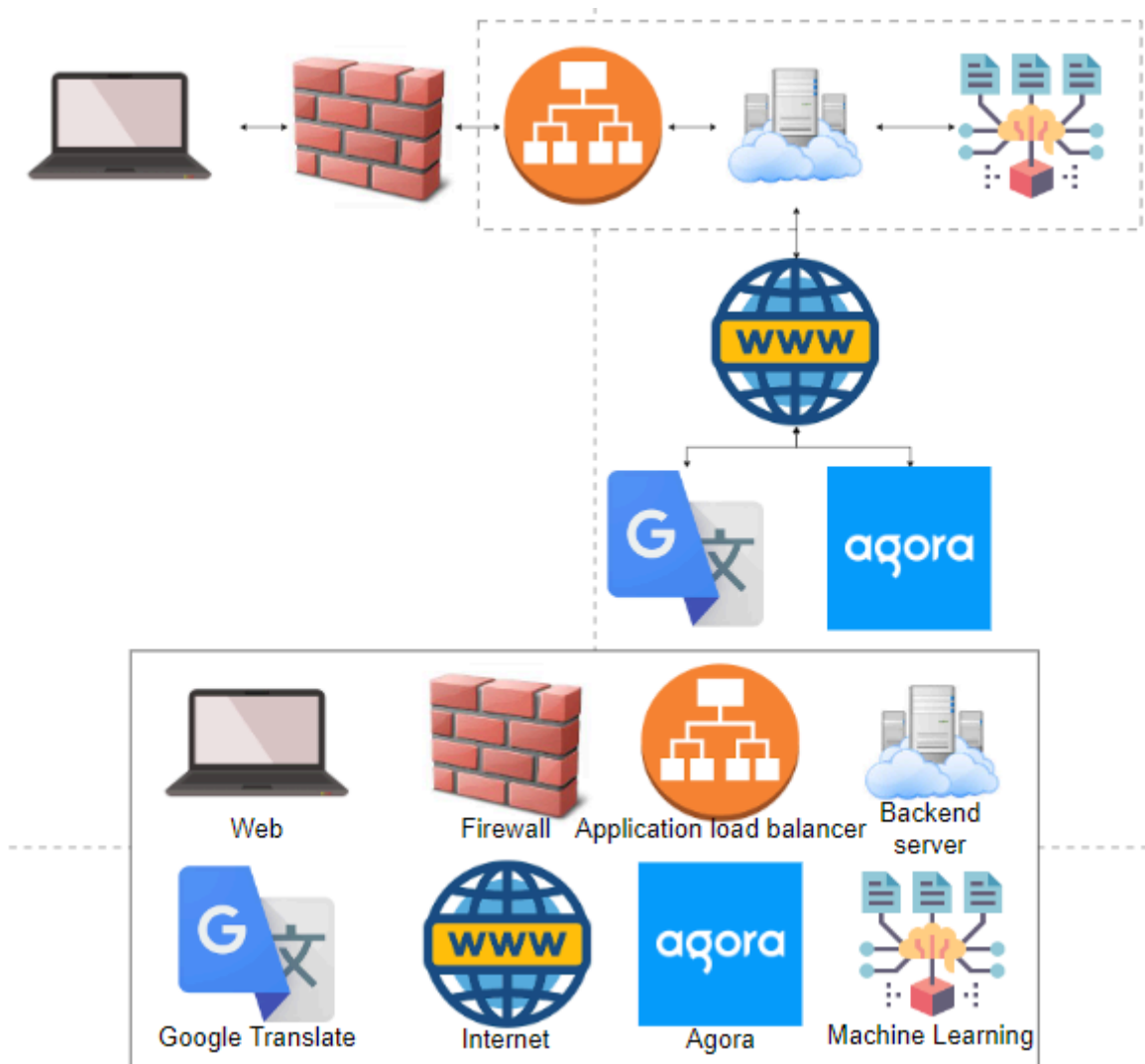


Figure 3: *Allocation View*

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Element	Responsibilities
Web	Admin, User access and use the system.
Firewall	Where to protect systems from cyber attacks and monitor and control data flows.
Application load balancer	Where load balancing between servers and traffic redirection is handled.
Backend server	The place to manage and maintain the database.
Google Translate	Integrate and maintain Google Translate API-based services.
Internet	Where to manage bandwidth and security.
Agora	Alert users when problems occur, minimizing risks as much as possible
Machine learning	Where to deploy and maintain streaming infrastructure and monitor and manage Agora's resource usage.

Description: A smart home system is a network of interconnected devices that can be controlled remotely using a smartphone or tablet app. The system typically consists of a central hub, a variety of smart devices, and a cloud-based service.

4. References

No.	References	Document Information
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		https://standards.ieee.org/standard/1471-2000.html
		https://ieeexplore.ieee.org/document/917550
2	Patterns	https://en.wikipedia.org/wiki/Architectural_pattern
3.	Evaluation standards	https://www.iso.org/obp/ui/#iso:std:iso-iec-ieee:42030:ed-1:v1:en
		https://gabrielfs7.github.io/software-architecture/2019/10/18/ata-m-analyze-evaluate-architecture/